

Water on Earth

From space, Earth looks blue. That's because water covers most of our planet. Water shapes our weather, carves our land, fills our oceans, and keeps every living thing alive. In this chapter you'll follow water as it moves all around the Earth.

 Georgia Standard S6E3

WHAT YOU'LL LEARN

01 Earth: The Water Planet

03 The Water Cycle

05 Waves & Currents

02 Where the Water Is

04 The Ocean Floor

06 Tides



3.1 Earth: The Water Planet

If you could shrink Earth and look at it, the first thing you'd notice is how much **blue** there is. About **71%** of Earth's surface is covered by water. That's why Earth is sometimes called the "water planet" or the "blue marble." No other planet we know of has so much liquid water — and that water is a big reason Earth can support life.

Why Water Matters

- Every living thing — plants, animals, and you — needs water to survive.
- Water shapes the land through **weathering** and **erosion** (you'll learn more in a later chapter).
- Water stores and moves heat, which drives our **weather** and **climate**.
- Oceans are home to a huge share of Earth's living things.

? Quick Check 3.1

1. About what percentage of Earth's surface is covered by water?
2. Give two reasons water is important to Earth.

3.2 Where the Water Is

Earth has a *lot* of water — but most of it is not the kind you can drink. Scientists sort water into **salt water** and **fresh water**, and the amounts are very uneven.

Where Is Earth's Water?

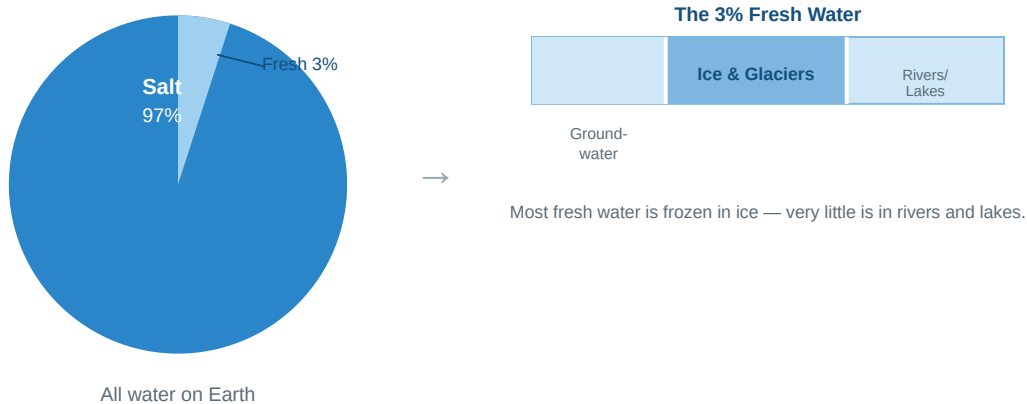


Figure 3.1 — Almost all of Earth's water is salty ocean water. Only a small slice is fresh, and most of that fresh water is frozen. (The fresh slice is drawn larger than 3% here so you can see it.)

Type / Location	About How Much	Notes
Oceans (salt water)	97% of all water	Too salty to drink or water crops
Ice caps & glaciers	About 2% of all water	Frozen fresh water, locked up at the poles and in mountains
Groundwater	Most of the rest	Fresh water stored underground in soil and rock (aquifers)
Rivers, lakes & swamps	A tiny fraction	The surface fresh water we most easily use

Aquifer: an underground layer of rock or soil that holds groundwater. Many towns pump drinking water up from aquifers using wells.

! Surprising but True

Even though Earth is covered in water, **less than 1%** of all the water is fresh water that people can easily reach and use. That's why protecting our rivers, lakes, and groundwater is so important.

? Quick Check 3.2

1. Is most of Earth's water salt water or fresh water?
2. Where is most of Earth's *fresh* water found?

3.3 The Water Cycle

Water doesn't stay in one place. It moves constantly between the oceans, the air, the land, and back again, in a never-ending loop called the **water cycle**. The energy that powers this whole cycle comes from one place: the **Sun**.

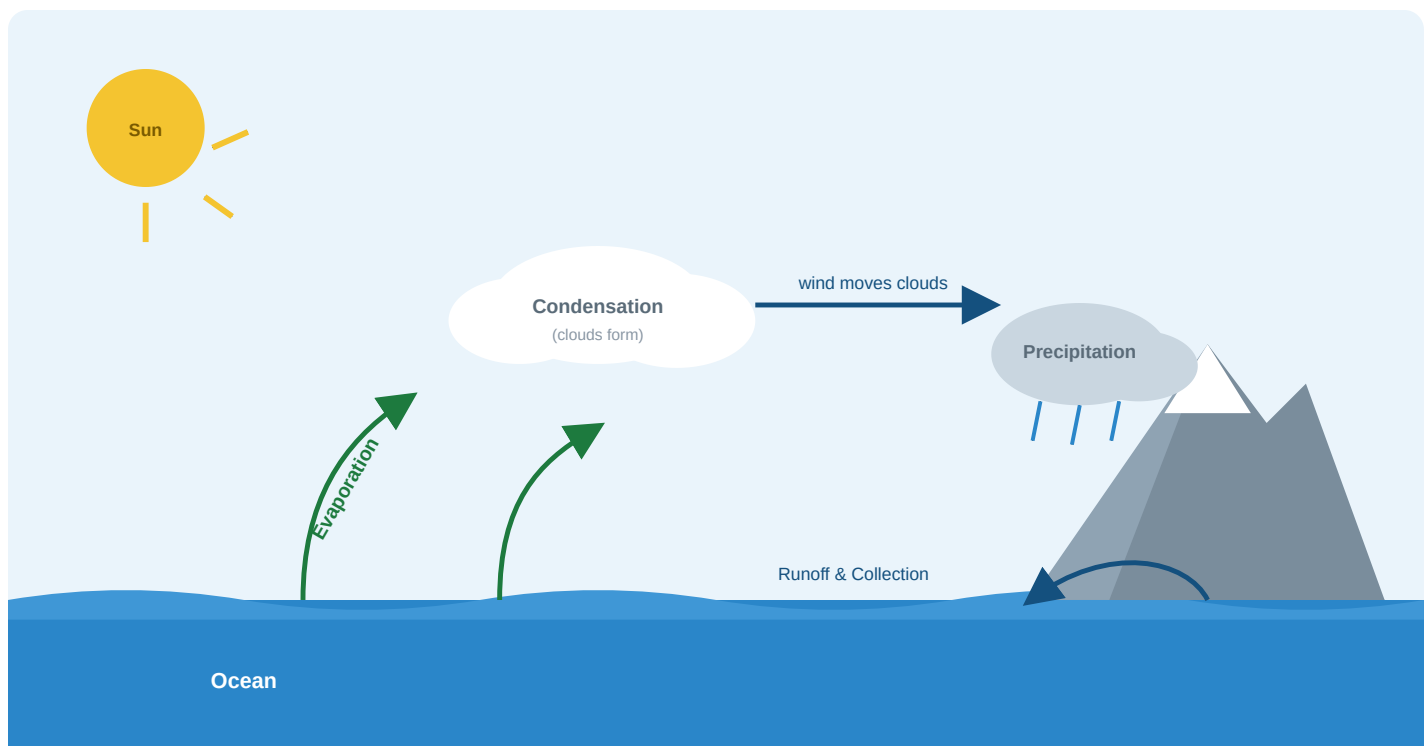


Figure 3.2 — The Sun's energy drives the water cycle. Water evaporates, condenses into clouds, falls as precipitation, and collects again.

The Main Steps

Step	What Happens	Driven By
Evaporation	The Sun heats water and turns it into invisible water vapor (a gas) that rises into the air	Heat from the Sun
Condensation	High up where it's cooler, water vapor turns back into tiny droplets, forming clouds	Cooling
Precipitation	Droplets join, grow heavy, and fall as rain, snow, sleet, or hail	Gravity
Collection / Runoff	Water collects in oceans, lakes, and rivers, or soaks into the ground — then the cycle repeats	Gravity

★ The Sun Is the Engine

Without the Sun's heat, water would never evaporate and the cycle would stop. The Sun lifts water into the sky; gravity brings it back down. Together they keep fresh water moving across the planet.

Transpiration: plants also release water vapor from their leaves. Along with evaporation, this adds water to the air. (Sometimes the two together are called "evapotranspiration.")

? Quick Check 3.3

1. What is the source of energy that drives the water cycle?
2. What is the difference between evaporation and condensation?
3. Name two forms that precipitation can take.

3.4 The Ocean Floor

The bottom of the ocean is not a flat, boring plain. Hidden under the water is a landscape with mountains taller than any on land, valleys deeper than any canyon, and wide flat stretches. Scientists map this hidden world using sound waves (called **sonar**) and satellites.

A Trip Across the Ocean Floor

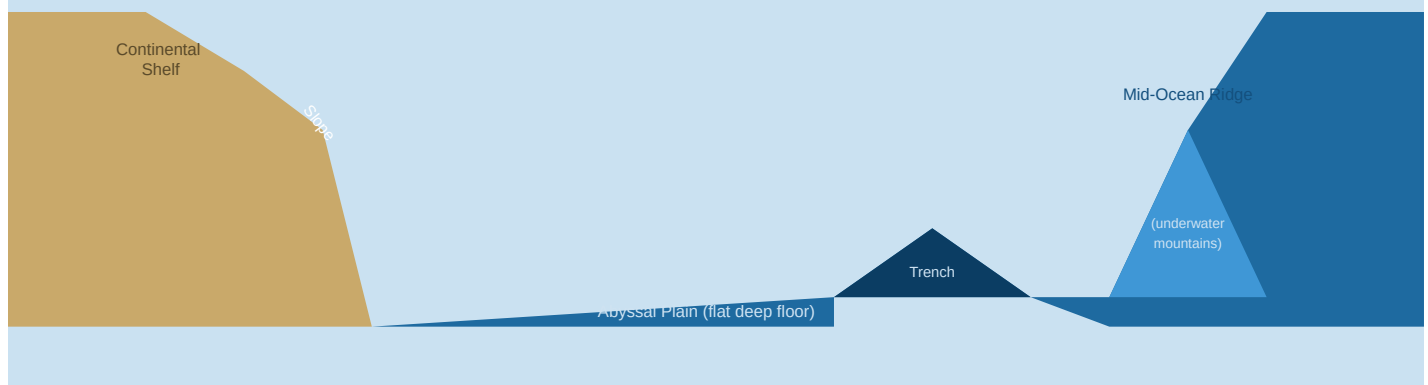


Figure 3.3 — A cross-section of the ocean floor, from the shallow edge of a continent to deep trenches and underwater mountain ranges.

Feature	What It Is
Continental Shelf	The shallow, gently sloping edge of a continent, just offshore
Continental Slope	The steeper drop-off where the shelf ends and the deep ocean begins
Abyssal Plain	The wide, flat floor of the deep ocean
Mid-Ocean Ridge	A chain of underwater mountains running along the sea floor
Trench	A deep, narrow valley — the deepest parts of the ocean

★ Deeper Than Everest Is Tall

The deepest ocean trench (the Mariana Trench) is about 11 kilometers deep. If you dropped Mount Everest into it, the mountain's peak would still be more than a kilometer underwater!

? Quick Check 3.4

1. What is the difference between the continental shelf and a trench?
2. What tool do scientists use to map the shape of the ocean floor?

3.5 Waves & Currents

Ocean water is always on the move. Two of the most important kinds of movement are **waves** and **currents** — and they are not the same thing.

Waves — energy moving through water

Most ocean **waves** are made by **wind** blowing across the water's surface. A wave carries **energy** forward, but the water itself mostly stays in place, bobbing up and down. Think of a wave moving across a stadium crowd — the people stand up and sit down, but they don't run around the stadium.

Currents — rivers within the ocean

A **current** is a steady flow of water moving in one direction, like a river inside the ocean. **Surface currents** are driven by wind. **Deep currents** are driven by differences in water temperature and saltiness — cold, salty water sinks, while warm water rises. Currents move **heat** around the planet, which affects climate.

	Wave	Current
What moves	Energy (water mostly stays put)	The water itself, in one direction
Main cause	Wind on the surface	Wind, plus temperature and saltiness
Effect	Shapes shorelines, makes surf	Moves heat and affects climate

? Quick Check 3.5

1. What causes most ocean waves?
2. How is a current different from a wave?
3. Why do currents matter for Earth's climate?

3.6 Tides

If you spend a day at the beach, you might notice the water creeping up the sand and later pulling back. This daily rise and fall of the ocean is called the **tides**, and it's caused mostly by the **Moon's gravity**.

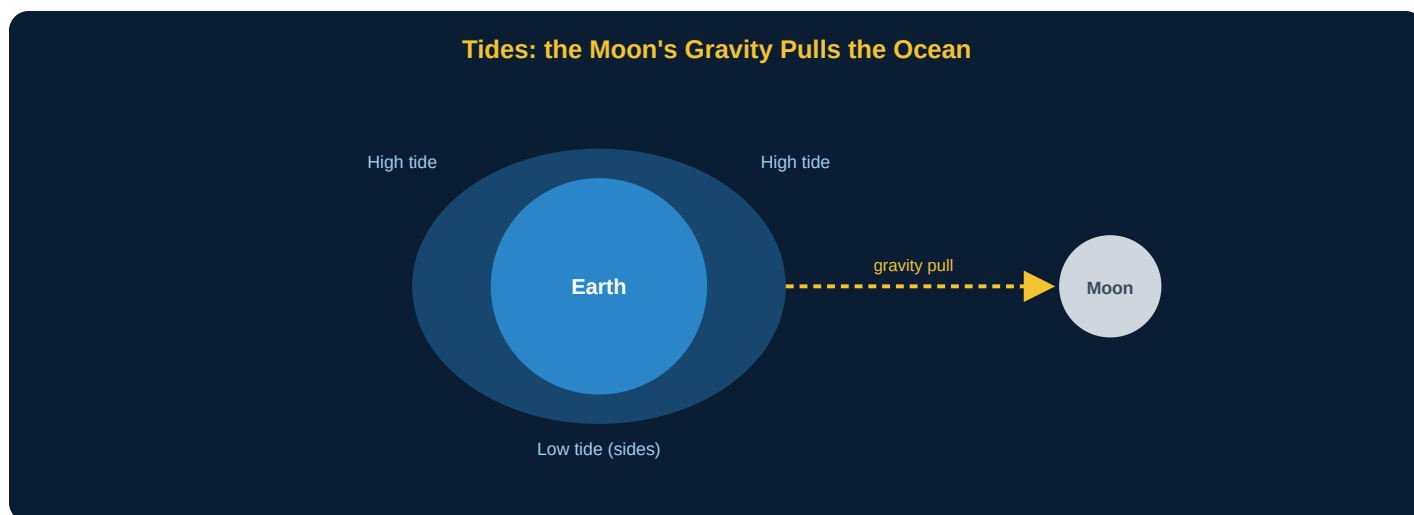


Figure 3.4 — The Moon's gravity pulls ocean water into bulges. Places under a bulge have high tide; places between them have low tide.

The Moon's gravity tugs on Earth's oceans, pulling the water into a **bulge** on the side facing the Moon (and another on the opposite side). Places under a bulge experience **high tide**. As Earth rotates, each coast passes through these bulges, so most beaches get **two high tides and two low tides** about every day.

★ The Sun Helps Too

The Sun's gravity also affects tides, but because the Sun is so far away, its pull is weaker than the Moon's. When the Sun and Moon line up, their pulls combine to make extra-large tides.

? Quick Check 3.6

1. What causes the tides?
2. About how many high tides does a typical beach get each day?

Chapter 3 Wrap-Up

★ Key Takeaways

- **Earth is a water planet** — about 71% of the surface is water, but most is salty ocean.
- **Fresh water is rare:** only about 3% of water is fresh, and most of that is frozen in ice and glaciers.
- **The water cycle** (evaporation, condensation, precipitation, collection) is powered by the Sun and gravity.
- **The ocean floor** has shelves, slopes, flat plains, ridges, and deep trenches.
- **Waves** carry energy; **currents** move water and heat. **Tides** are the daily rise and fall caused by the Moon's gravity.